

Consumption, Saving, and Investment

ECON202 MACROECONOMICS-1

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April 17, 2026

Chapter Outline

- Describe the factors that affect consumption and saving decisions
(Sec. 4.1: Consumption and Saving)
- Discuss the factors that affect the investment of firms
(Sec. 4.2: Investment)
- Explain the factors affecting goods market equilibrium
(Sec. 4.3: Goods Market Equilibrium)

Roadmap of Lecture

Consumption and Saving

Investment

Summary

Diwali Greetings

The importance of consumption and saving

- Desired consumption: consumption amount desired by households
- Desired national saving: level of national saving when consumption is at its desired level:

$$S^d = Y - C^d - G \quad (4.1) \tag{1}$$

Consumption and Saving

The consumption and saving decision of an individual

- A person can consume less than current income (saving is positive)
- A person can consume more than current income (saving is negative)

The consumption and saving decision of an individual

Trade-off between current consumption and future consumption

- The price of 1 unit of current consumption is $1 + r$ units of future consumption, where r is the real interest rate
- Consumption-smoothing motive: the desire to have a relatively even pattern of consumption over time

Effect of changes in current income

- Increase in current income: both consumption and saving increase (vice versa for decrease in current income)
- Marginal propensity to consume (MPC) = fraction of additional current income consumed in current period; between 0 and 1
- Aggregate level: When current income (Y) rises, C^d rises, but not by as much as Y , so S^d rises

Effect of changes in expected future income

- Higher expected future income leads to more consumption today, so saving falls

Keynesian Consumption Function

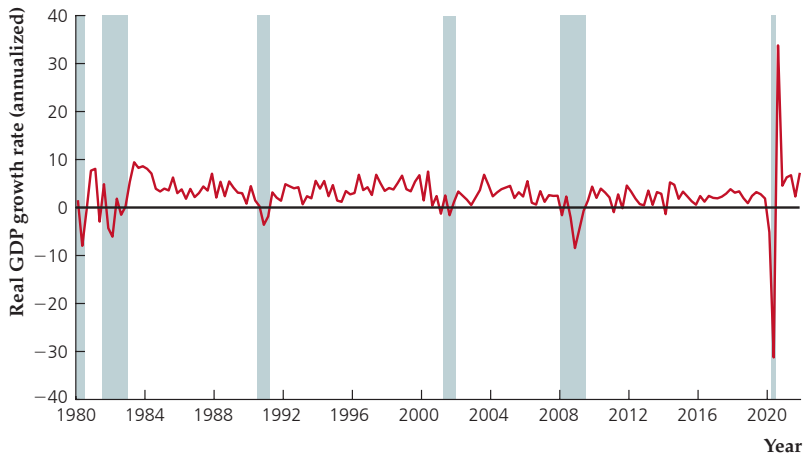
The Keynesian consumption function is given by:

$$C^d = c_0 + c_y Y \quad (2)$$

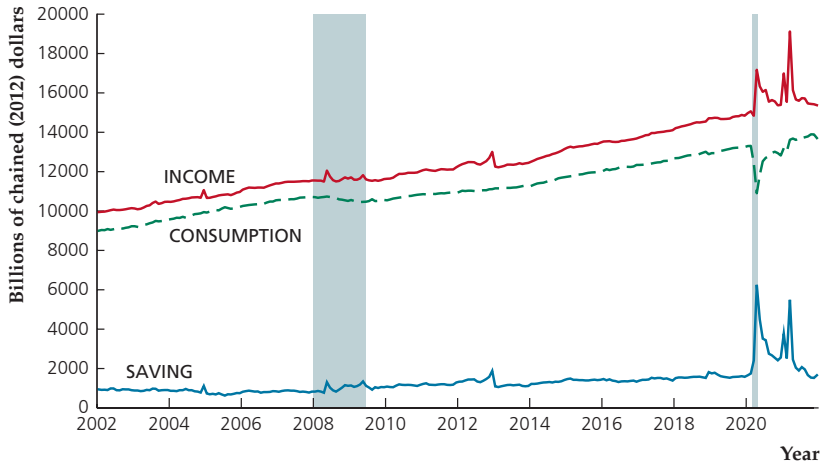
Where:

- C^d = Total consumption spending
- c_0 = Autonomous consumption (consumption when income is zero)
- c_y = Marginal propensity to consume (MPC), where $0 < c < 1$
- Y = Disposable income

Application: impact of the pandemic recession on Consumption



Application: Income, Consumption, and Saving



Effect of changes in wealth

- Increase in wealth raises current consumption, so lowers current saving

Effect of changes in real interest rate

increased real interest rate has two opposing effects

- **Substitution effect:** Positive effect on saving, since rate of return is higher; greater reward for saving elicits more saving
- **Income effect**
 - For a saver: Negative effect on saving, since it takes less saving to obtain a given amount in the future (target saving)
 - For a borrower: Positive effect on saving, since the higher real interest rate means a loss of wealth
- Empirical studies have mixed results; probably a slight increase in aggregate saving

Consumption and Saving

Effect of changes in real interest rate: Taxes and the real return to saving

- Expected after-tax real interest rate:

$$r_{a-t} = (1 - t)i - \pi^e \quad (4.2) \quad (3)$$

Table 4.1 Calculating After-Tax Interest Rates

i = nominal interest rate = 5% per year

π^e = expected inflation rate = 2% per year

Example 1

t = tax rate on interest income = 30%

Nominal after-tax interest rate = $(1 - t)i = (1 - 0.30)5\% = 3.5\%$

Expected real after-tax interest rate = $(1 - t)i - \pi^e = (1 - 0.30)5\% - 2\% = 1.5\%$

Example 2

t = tax rate on interest income = 20%

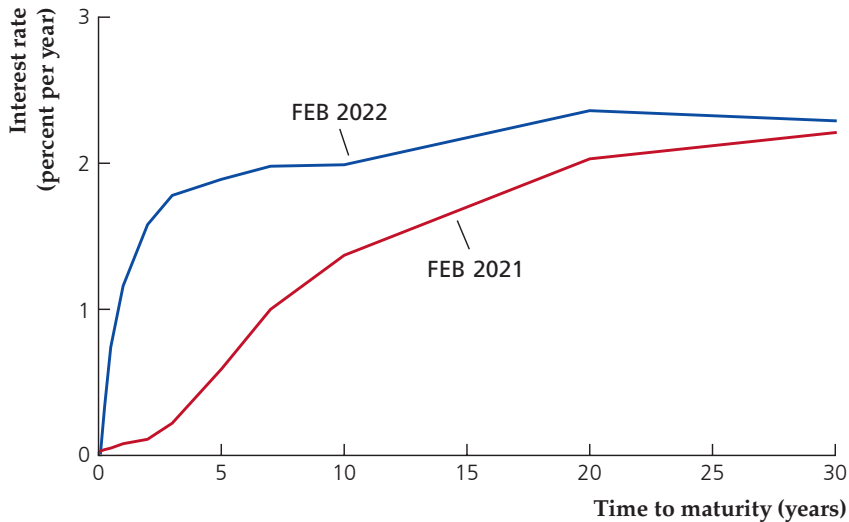
Nominal after-tax interest rate = $(1 - t)i = (1 - 0.20)5\% = 4\%$

Expected real after-tax interest rate = $(1 - t)i - \pi^e = (1 - 0.20)5\% - 2\% = 2\%$

In touch with data and research: interest rates

- different interest rates, default risk, term structure (yield curve), and tax status
- Since interest rates often move together, we frequently refer to "the" interest rate
- Yield curve: relationship between life of a bond and the interest rate it pays

Yield Curve



Consumption and Saving

Fiscal policy

- Affects desired consumption through changes in current and expected future income
- Directly affects desired national saving: $S^d = Y - C^d - G$

Fiscal policy: Government purchases (temporary increase)

- Higher G financed by higher current taxes reduces after-tax income, lowering desired consumption
- Even true if financed by higher future taxes, if people realize how future incomes are affected
- So government purchases reduce both C^d and S^d

Taxes

- Lump-sum tax cut today, financed by higher future taxes
- Decline in future income may offset increase in current income; desired consumption could rise or fall

Taxes

Ricardian equivalence proposition

- If future income loss exactly offsets current income gain, no change in consumption
- Tax change affects only the timing of taxes, not their ultimate amount (PV)
- In practice, people may not see that future taxes will rise if taxes are cut today; then a tax cut leads to increased desired consumption and reduced desired national saving

Summary 5: Determinants of Desired National Saving

An increase in	Causes desired national saving to	Reason
Current output, Y	Rise	Part of the extra income is saved to provide for future consumption.
Expected future output	Fall	Anticipation of future income raises current desired consumption, lowering current desired saving.
Wealth	Fall	Some of the extra wealth is consumed, which reduces saving for given income.
Expected real interest rate, r	Probably rise	An increased return makes saving more attractive, probably outweighing the fact that less must be saved to reach a specific savings target.
Government purchases, G	Fall	Higher government purchases directly lower desired national saving.
Taxes, T	Remain unchanged or rise	Saving doesn't change if consumers take into account an offsetting future tax cut; saving rises if consumers don't take into account a future tax cut and thus reduce current consumption.

Roadmap of Lecture

Consumption and Saving

Investment

Summary

Diwali Greetings

Why is investment important?

- Investment fluctuates sharply over the business cycle, so we need to understand investment to understand the business cycle
- Investment plays a crucial role in economic growth

The desired capital stock

- Desired capital stock is the amount of capital that allows firms to earn the largest expected profit
- Desired capital stock depends on costs and benefits of additional capital
- Since investment becomes capital stock with a lag, the benefit of investment is the future marginal product of capital (MPK^f)

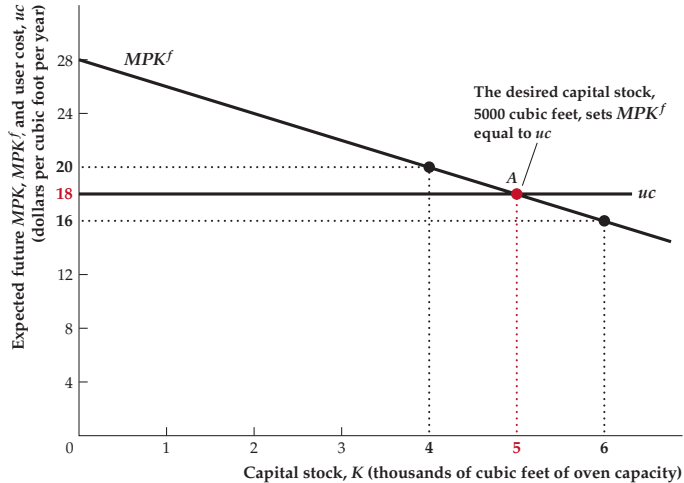
The desired capital stock

The user cost of capital

- Example of Kyle's Bakery: cost of capital, depreciation rate, and expected real interest rate
- User cost of capital = real cost of using a unit of capital for a specified period of time
= real interest cost + depreciation

$$uc = rp_K + dp_K = (r + d)p_K \quad (4.3) \quad (4)$$

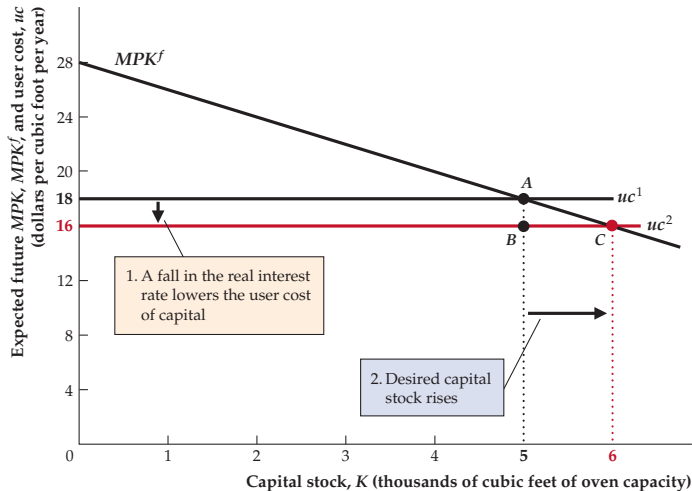
Determination of the desired capital stock



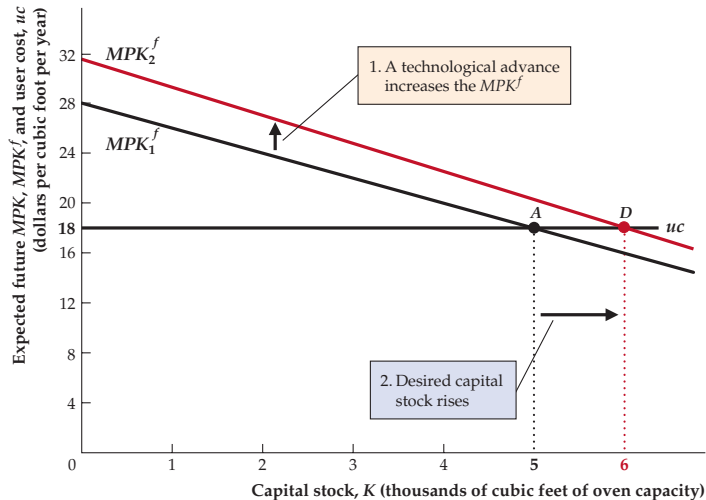
Factors

- Factors that shift the MPK^f curve or change the user cost of capital cause the desired capital stock to change
- These factors are changes in the real interest rate, depreciation rate, price of capital, or technological changes that affect the MPK^f

Decline in the real interest rate raises the desired capital stock



An increase in the expected MPK^f raises the desired capital stock



Taxes and the desired capital stock

- With taxes, the return to capital is only $(1 - \tau)MPK^f$
- A firm chooses its desired capital stock so that the return equals the user cost, so $(1 - \tau)MPK^f = uc$, which means:

$$MPK^f = \frac{uc}{1 - \tau} = \frac{(r + d)p_K}{1 - \tau} \quad (4.4)$$

Taxes and the desired capital stock

- Tax-adjusted user cost of capital is $\frac{uc}{1-\tau}$
- An increase in τ raises the tax-adjusted user cost and reduces the desired capital stock

From the desired capital stock to investment

The capital stock changes from two opposing channels

- New capital increases the capital stock; this is gross investment
- The capital stock depreciates, which reduces the capital stock

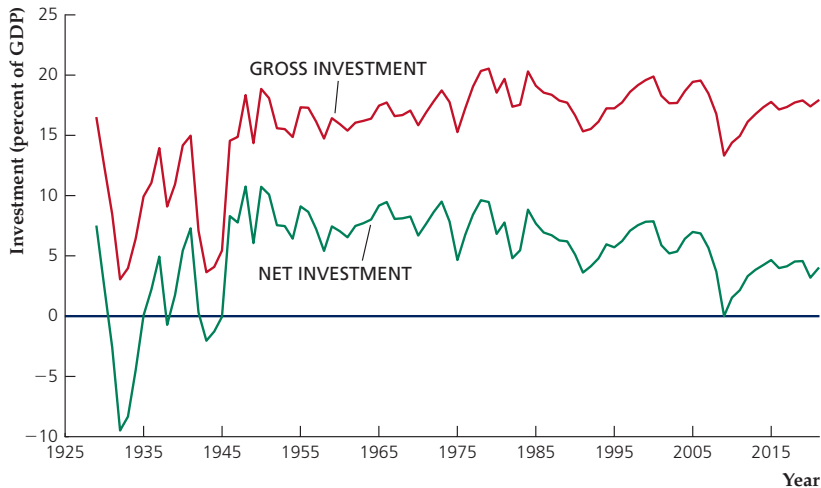
From the desired capital stock to investment

- Net investment = gross investment (I) minus depreciation:

$$K_{t+1} - K_t = I_t - dK_t \quad (4.5) \quad (5)$$

- where net investment equals the change in the capital stock

Figure 4.6 Gross and net investment, 1929-2011



From the desired capital stock to investment

- Rewriting (4.5) gives $I_t = K_{t+1} - K_t + dK_t$
- If firms can change their capital stocks in one period, then the desired capital stock $(K^*) = K_{t+1}$
- So $I_t = K^* - K_t + dK_t$ (4.6)

Thus investment has two parts

- Desired net increase in the capital stock over the year ($K^* - K_t$)
- Investment needed to replace depreciated capital (dK_t)

Lags and investment

- Some capital can be constructed easily, but other capital may take years to put in place
- So investment needed to reach the desired capital stock may be spread out over several years

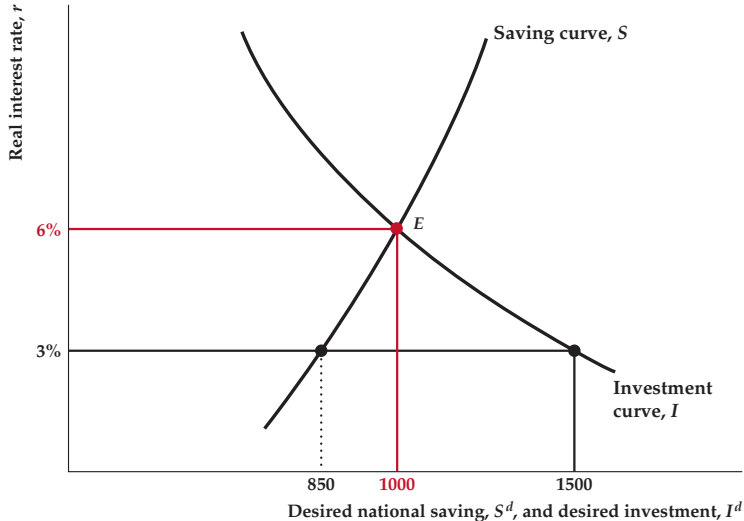
Summary 6

Determinants of Desired Investment		
An increase in	Causes desired investment to	Reason
Real interest rate, r	Fall	The user cost increases, which reduces desired capital stock.
Effective tax rate, τ	Fall	The tax-adjusted user cost increases, which reduces desired capital stock.
Expected future MPK	Rise	The desired capital stock increases.

goods market equilibrium condition

- $Y = C^d + I^d + G$ (4.7)
- Differs from income-expenditure identity, as goods market equilibrium condition need not hold
- undesired goods may be produced, so goods market won't be in equilibrium
- Alternatively, using relationship between S^d and I^d .
 - $S^d = Y - C^d - G,$
 - $S^d = I^d$ (4.8)

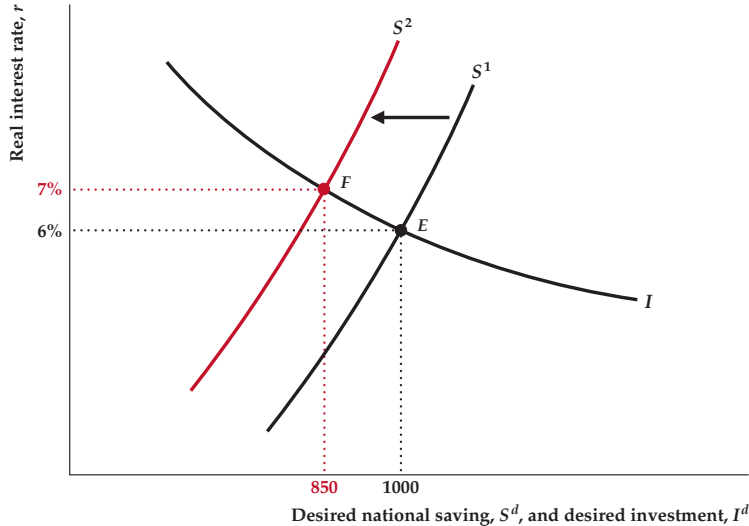
Goods market equilibrium: Plot S^d vs. I^d



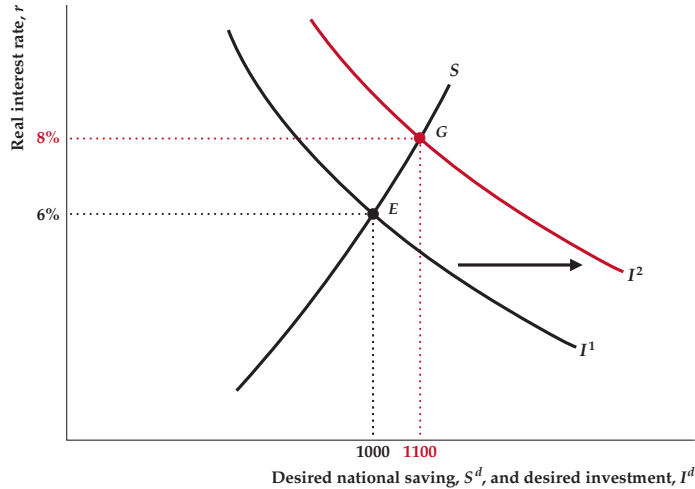
Components of Aggregate Demand for Goods (An Example)

Variable	$r = 3\%$	$r = 6\%$
Output, Y	4500	4500
Desired Consumption, C^d	2150	2000
Desired Investment, I^d	1500	1000
Government Purchases, G	1500	1500
Desired National Saving, $S^d = Y - C^d - G$	850	1000
Aggregate Demand, $C^d + I^d + G$	5150	4500

A decline in desired saving



An increase in desired investment



Roadmap of Lecture

Consumption and Saving

Investment

Summary

Diwali Greetings

1. The Saving-Consumption Decision

- **Saving** equals income minus consumption; the consumption and saving decisions are inherently linked
- Households save because they value both current and future consumption
- Higher current saving reduces current consumption but increases future consumption possibilities
- The saving decision reflects a trade-off between present and future consumption

2. Income, Wealth, and Saving

- At both household and national levels, an increase in **current income** raises both desired consumption and desired saving
- An increase in **expected future income** or in **wealth** raises desired consumption but reduces current desired saving
- An increase in wealth raises desired consumption and reduces desired saving
- These changes affect consumption without changing current income or output

3. Real Interest Rate Effects on Saving (Part 1)

- **Substitution effect:** Higher r increases the relative price of current consumption (costs $1 + r$ units of future consumption), encouraging more saving
- **Income effect:** Higher r makes savers wealthier (more interest income) but borrowers poorer (more interest payments)
- For savers: substitution effect (increases saving) and income effect (decreases saving) work in opposite directions

3. Real Interest Rate Effects on Saving (Part 2)

- For borrowers: both substitution and income effects increase saving when r rises
- Empirically, an increase in the real interest rate increases desired national saving and reduces consumption, but the effects are small
- The relevant rate is the **expected real after-tax interest rate**

4. Government Purchases and Saving

- With output constant, a **temporary increase in government purchases** reduces desired consumption
- Higher G implies higher present or future taxes, making consumers feel poorer
- The decrease in C^d is smaller than the increase in G
- Therefore, **desired national saving** ($S^d = Y - C^d - G$) falls

5. Ricardian Equivalence

- The **Ricardian equivalence proposition** states that a current lump-sum tax cut has no effect on desired consumption or national saving
- Logic: if government purchases are unchanged, a tax cut that increases current income must be offset by future tax increases
- Higher current income is exactly offset by lower expected future income
- If consumers ignore future tax changes, Ricardian equivalence fails and a tax cut raises C^d and lowers S^d

6. Desired Capital Stock

- The **desired capital stock** maximizes expected profits
- At the optimum: expected future marginal product of capital (MPK^f) equals **user cost of capital**
- User cost = depreciation cost + interest cost = $(\delta + r) \times P_K$
- This represents the expected real cost of using one unit of capital for a period

7. Determinants of Desired Capital Stock

- Desired capital stock increases when:
 - **User cost of capital** decreases
 - **Expected future marginal product of capital** increases
 - **Effective tax rate** on capital decreases
- These changes make capital investment more profitable

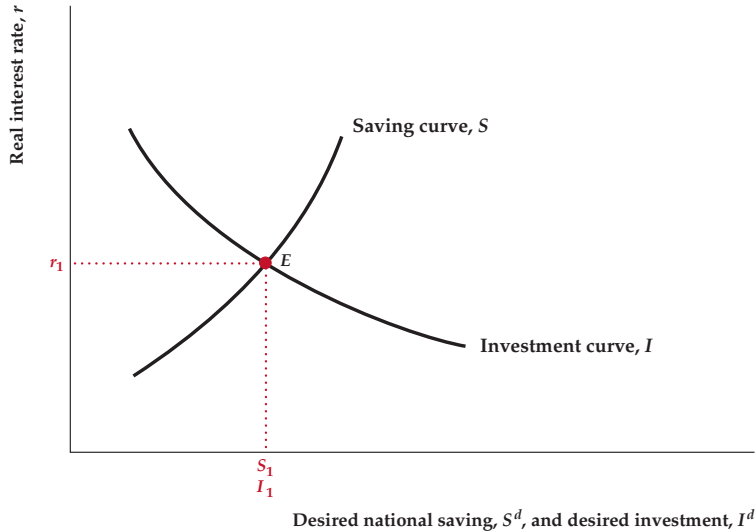
8. Investment and Capital Stock

- **Gross investment** is spending on new capital goods
- **Net investment** = Gross investment – Depreciation = ΔK
- Firms invest to achieve their desired capital stock level
- When desired capital stock increases, firms invest more

9. Goods Market Equilibrium

- **Goods market equilibrium:** aggregate quantity supplied equals aggregate quantity demanded
- In a closed economy: $Y = C^d + I^d + G$
- Equivalently: desired national saving equals desired investment ($S^d = I^d$)
- For any given output level, the **real interest rate** adjusts to bring the goods market into equilibrium

10. Saving-Investment Diagram



Roadmap of Lecture

Consumption and Saving

Investment

Summary

Diwali Greetings

Dear Students,

$Y = C + I + G + NX$... but this Diwali,

$Y = \text{Yummy food} + \text{Infinite joy} + \text{Glowing diyas} + \text{Net eXplosive fun!}$

May your happiness production function be: $H = A(\text{Sweets})^{0.5}(\text{Lights})^{0.5}$